

6	(a)		Faraday's Law of Electromagnetic Induction states that the <u>induced e.m.f is directly proportional to the rate of change of magnetic flux linkage</u> . [B1]
	(b)	(i)	At $t = 0$ and $t = 0.25$ s $B = 1.16 \times 10^{-2}$ T and $B = 1.24 \times 10^{-2}$ T respectively [C1] Average induced e.m.f $= NA \left(\frac{\Delta B}{t} \right)$ $= 240 (2.5 \times 10^{-4}) \left[\frac{(1.24 - 1.16) \times 10^{-2}}{0.25} \right]$ [C1] $= 1.9 \times 10^{-4}$ V [A1]
		(ii)	<u>Gradient at the steepest parts of the graph</u> ($t = 0, 0.5, 1.0$) in Fig. 8.2 [B1]
		(iii)	Correct shape: Negative cosine graph (also accept cosine graph) [B1] Zero e.m.f. at $t = 0.25$ and 0.75 s, max e.m.f at $t = 0, 5.0$ and 1.0 s [B1]
		(iv)	<u>An induced current</u> now flows through the coil [B1] By Lenz's Law, a magnetic field will be generated through the coil that exerts <u>a magnetic force on the magnet to oppose its motion</u> [M1] Hence the magnet experiences damping and the <u>oscillation dies down / amplitude of oscillation decreases/ gradually comes to a stop</u> [A1] Or <u>An induced current</u> now flows through the coil [B1] Since current is induced, it implies that part of the total <u>mechanical energy of the system is converted into electrical energy and dissipated as heat</u> through the resistor [M1] The process continues until all the initial total mechanical energy of the magnet decreases to zero, i.e. the magnet experiences damping and the <u>oscillation dies down / amplitude of oscillation decreases/ gradually comes to a stop</u> [A1]

7	(a)	(i)	Battery capacity	
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